# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 6 | Common prime factors give HCF $= 6$ |
| 2 | **(d)** 6 | Substitute $x = 4$: $p(4) = 6$ |
| 3 | **(a)** infinitely many solutions (coincident lines) | $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{1}{2}$ |
| 4 | **(a)** $x^2 + 3x + 2 = 0$ | Degree must be 2 |
| 5 | **(c)** 12 | LCM $= \dfrac{4 \times 6}{2} = 12$ |
| 6 | **(c)** 4 | $2x - 8 = 0 \Rightarrow x = 4$ |
| 7 | **(a)** $(3, 2)$ | $2(3) + 3(2) = 12 = 12$ |
| 8 | **(a)** 2 | At $x = 2$ the left side equals $0$, so 2 is a root |
| 9 | **(b)** 3 | $12 = 2^2 \times 3$ |
| 10 | **(b)** 5 | Sum of zeroes $= -b/a$ |
| 11 | **(c)** 7 | $k = 2(2) + 3(1)$ |
| 12 | **(b)** -3 | Compare with $ax^2 + bx + c = 0$ |
| 13 | **(b)** $\sqrt{7}$ | 7 is not a perfect square, so $\sqrt{7}$ is irrational |
| 14 | **(b)** 6 | Product of zeroes $= c/a$ |
| 15 | **(a)** 7 | $x + y = 10$, $x - y = 4$ gives $2x = 14$ |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $18 = 2 \times 3^2$, $24 = 2^3 \times 3$ (1); HCF $= 6$ (1). | 1+1 |
| 17 | $p(3) = 10$ (2). | 2 |
| 18 | $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$ (1); all ratios equal, infinitely many solutions, consistent (1). | 1+1 |
| 19 | Put $x = 2$: LHS $= 0$ (1); equals RHS 0, so it is a root (1). | 1+1 |
| 20 | Divide repeatedly by primes (1); $140 = 2^2 \times 5 \times 7$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $26 = 2 \times 13$, $91 = 7 \times 13$ (1); HCF $= 13$ (1); LCM $= 182$ (1). | 1+1+1 |
| 22 | Factors $(x - 3)(x - 5)$ (1); zeroes $3, 5$ (1); sum $8 = -b/a$ verified (1). | 1+1+1 |
| 23 | Multiply (i) by $-1$ and (ii) by 1 and subtract: $(-2)x = -10$ (2); $x = 5$ (1); put in (i): $y = 3$ (1); check in (ii) (1). | 2+1+1+1 |
| 24 | Split the middle term: $-5 = -3 - 2$ because $ac = 6$ (1); regroup and factorise (1); roots $x = 2, 3$ (1). | 1+1+1 |
| | **OR:** HCF $= 6$, LCM $= 60$ (1+1); HCF×LCM $= 360 = 12 \times 30$ (1). | 1+1+1 |
| 25 | $p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (5)x + (6)$ (1); one such polynomial is $x^2 - 5x + 6$ (1). | 1+1+1 |
| | **OR:** From (i): $y = \dfrac{18 - 2x}{3}$ (1); substitute in (ii) (1); simplify to $(1)x = 3$ (1); $x = 3$ (1); $y = 4$ (1). | 1+1+1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $135 = 3^3 \times 5$, $225 = 3^2 \times 5^2$ (2); HCF $= 45$ (1); LCM $= 675$ (1); $45 \times 675 = 30375$ verified (1). | 2+1+1+1 |
| | **OR:** Zeroes 2, 3 (2); sum $= 5 = -\dfrac{b}{a}$ (1.5); product $= 6 = \dfrac{c}{a}$ (1.5). | 2+1.5+1.5 |
| 27 | Multiply (i) by 4 and (ii) by 3 and subtract: $(-7)x = -21$ (2); $x = 3$ (1); put in (i): $y = 2$ (1); check in (ii) (1). | 2+1+1+1 |
| | **OR:** Split the middle term: $-7 = -6 - 1$ because $ac = 6$ (1); regroup and factorise (1); roots $x = \frac{1}{2}, 3$ (1). | 1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 2 and 3 | Put each factor equal to zero |
| 2 | **(b)** $2^2 \times 3 \times 5$ | Divide repeatedly by primes: $60 = 2^2 \times 3 \times 5$ |
| 3 | **(d)** 2 | A polynomial of degree $n$ has at most $n$ zeroes |
| 4 | **(b)** $y$-axis | All points have the same $x$-coordinate |
| 5 | **(d)** 1 | $D = b^2 - 4ac$ |
| 6 | **(a)** terminating | Denominator $8$: only 2 and 5 as prime factors |
| 7 | **(a)** 2 | $p(2) = 0 = 0$, while the other values do not make $p(x)$ zero |
| 8 | **(a)** coincident | One equation is a multiple of the other |
| 9 | **(a)** two distinct real roots | $D = b^2 - 4ac = 1$ |
| 10 | **(b)** 10 | Common prime factors give HCF $= 10$ |
| 11 | **(b)** 16 | Substitute $x = 3$: $p(3) = 16$ |
| 12 | **(a)** no solution (parallel lines) | $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$, $\frac{c_1}{c_2} = \frac{5}{7}$ |
| 13 | **(c)** $x^2 - 5x + 6 = 0$ | Degree must be 2 |
| 14 | **(c)** 18 | LCM $= \dfrac{6 \times 9}{3} = 18$ |
| 15 | **(a)** -3 | $3x + 9 = 0 \Rightarrow x = -3$ |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $3x - 12 = 0$ (1) so $x = 4$ (1). | 1+1 |
| 17 | Adding: $2x = 16$ (1) so $x = 8$, $y = 4$ (1). | 1+1 |
| 18 | $x - 4 = 0$ or $x - 5 = 0$ (1); $x = 4$ or $x = 5$ (1). | 1+1 |
| 19 | $16 = 2^4$ (1); it has only 2 and/or 5, so terminating (1). | 1+1 |
| 20 | Factorise: $(x - 2)(x - 5)$ (1); zeroes 2 and 5 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $a = 2$, $b = -4$, $c = 3$ (1); $D = -8$ (1); no real roots (1). | 1+1+1 |
| 22 | $16 = 2^4$ (1); terminating (1); value $\approx 0.1875$ (1). | 1+1+1 |
| 23 | Put $x = 1$: $p(1) = 0$ (2) $= 0$, hence a zero (1). | 2+1 |
| 24 | Let numbers be $x, y$: $x + y = 25$, $x - y = 11$ (1); adding $2x = 36$ (1); numbers 18 and 7 (1). | 1+1+1 |
| | **OR:** Let integers be $x$ and $x + 1$: $x(x + 1) = 20$ (1); $x^2 + x - 20 = 0$ (1); $(x - 4)(x + 5) = 0$, $x = 4$ (1); integers 4 and 5 (1). | 1+1+1+1 |
| 25 | $96 = 2^5 \times 3$, $404 = 2^2 \times 101$ (1); HCF $= 4$ (1); LCM $= 9696$ (1). | 1+1+1 |
| | **OR:** Factors $(x + 2)(x - 6)$ (1); zeroes $-2, 6$ (1); sum $4 = -b/a$ verified (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $24 = 2^3 \times 3$ (3); HCF $= 6$ (1); LCM $= 72$ (1). | 3+1+1 |
| | **OR:** Sum $= 6 = -\dfrac{b}{a}$ (2); sum of products in pairs $= 11 = \dfrac{c}{a}$ (2); product $= 6 = -\dfrac{d}{a}$ (1). | 2+2+1 |
| 27 | Let pencil $= x$, eraser $= y$: $2x + 3y = 22$, $3x + 2y = 23$ (2); solving, $x = 5$, $y = 4$ (2); pencil ₹5, eraser ₹4 (1). | 2+2+1 |
| | **OR:** Breadth $x$, length $x + 5$: $x(x + 5) = 66$ (1); $x^2 + 5x - 66 = 0$ (1); $(x - 6)(x + 11) = 0$ (1); $x = 6$ m (breadth), length 11 m (reject negative) (2). | 1+1+1+2 |

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10

Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** $(2, 5)$ | $3(2) + 1(5) = 11 = 11$ |
| 2 | **(d)** 3 | At $x = 3$ the left side equals $0$, so $3$ is a root |
| 3 | **(c)** 3 | $30 = 2 \times 3 \times 5$ |
| 4 | **(d)** 4 | Sum of zeroes $= -b/a$ |
| 5 | **(c)** 14 | $k = 3(4) + 1(2)$ |
| 6 | **(a)** 7 | Compare with $ax^2 + bx + c = 0$ |
| 7 | **(d)** $\sqrt{11}$ | $11$ is not a perfect square, so $\sqrt{11}$ is irrational |
| 8 | **(a)** -5 | Product of zeroes $= c/a$ |
| 9 | **(b)** 10 | $x + y = 14$, $x - y = 6$ gives $2x = 20$ |
| 10 | **(b)** -1 and 4 | Put each factor equal to zero |
| 11 | **(a)** $2^2 \times 3 \times 7$ | Divide repeatedly by primes: $84 = 2^2 \times 3 \times 7$ |
| 12 | **(c)** 3 | A polynomial of degree $n$ has at most $n$ zeroes |
| 13 | **(a)** $y$-axis | All points have the same $x$-coordinate |
| 14 | **(b)** 1 | $D = b^2 - 4ac$ |
| 15 | **(a)** non-terminating repeating | Denominator $15$: has a prime factor other than 2 and 5 |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | $k = 3(2) + 2(5)$ (1) $= 16$ (1). | 1+1 |
| 17 | $D = (-6)^2 - 4(1)(8) = 4$ (1); two distinct real roots (1). | 1+1 |
| 18 | $30 = 2 \times 3 \times 5$, $45 = 3^2 \times 5$ (1); HCF $= 15$ (1). | 1+1 |
| 19 | $p(-2) = 0$ (2). | 2 |
| 20 | $\frac{a_1}{a_2} = \frac{1}{2}$, $\frac{b_1}{b_2} = \frac{1}{2}$ (1); $a_1/a_2 = b_1/b_2 \neq c_1/c_2$, no solution, inconsistent (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Multiply (i) by $-1$ and (ii) by 1 and subtract: $(-3)x = -12$ (2); $x = 4$ (1); put in (i): $y = 2$ (1); check in (ii) (1). | 2+1+1+1 |
| 22 | Split the middle term: $-4 = -5 + 1$ because $ac = -5$ (1); regroup and factorise (1); roots $x = -1, 5$ (1). | 1+1+1 |
| 23 | HCF $= 7$, LCM $= 70$ (1+1); HCF×LCM $= 490 = 14 \times 35$ (1). | 1+1+1 |
| 24 | $p(x) = k[x^2 - (\text{sum})x + \text{product}]$ (1); $x^2 - (-3)x + (2)$ (1); one such polynomial is $x^2 + 3x + 2$ (1). | 1+1+1 |
| | **OR:** From (i): $y = \dfrac{12 - x}{3}$ (1); substitute in (ii) (1); simplify to $(5)x = 30$ (1); $x = 6$ (1); $y = 2$ (1). | 1+1+1+1+1 |
| 25 | $a = 1$, $b = -8$, $c = 16$ (1); $D = 0$ (1); two equal real roots (1). | 1+1+1 |
| | **OR:** $25 = 5^2$ (1); terminating (1); value $\approx 0.28$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $196 = 2^2 \times 7^2$, $280 = 2^3 \times 5 \times 7$ (2); HCF $= 28$ (1); LCM $= 1960$ (1); $28 \times 1960 = 54880$ verified (1). | 2+1+1+1 |
| | **OR:** Zeroes $-4$, $1$ (2); sum $= -3 = -\dfrac{b}{a}$ (1.5); product $= -4 = \dfrac{c}{a}$ (1.5). | 2+1.5+1.5 |
| 27 | Multiply (i) by $-3$ and (ii) by 2 and subtract: $(-13)x = -52$ (2); $x = 4$ (1); put in (i): $y = 1$ (1); check in (ii) (1). | 2+1+1+1 |
| | **OR:** Split the middle term: $-1 = -3 + 2$ because $ac = -6$ (1); regroup and factorise (1); roots $x = -\frac{2}{3}, 1$ (1). | 1+1+1 |

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